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Problem 1 Write the correct answers in the table in the last page.

(22 points)

1- What types of impurity atoms are added to increase the number of conduction-band electrons in intrinsic silicon?

- a- bivalent b- octavalent c- pentavalent d- trivalent e- none of the above

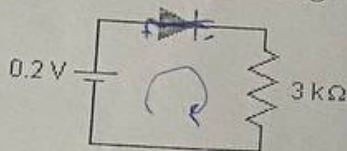
2- What does a high resistance reading in both forward- and reverse-bias directions indicate?

- a. A good diode b) damage diode c) A short circuit diode d) A defective ohmmeter

3- A silicon practical diode is forward-biased. You measure the voltage to ground from the anode at _____, and the voltage from the cathode to ground at _____.

- a- 0 V, 0.3 V b- 2.3 V, 1.6 V c- 1.6 V, 2.3 V d- 0.3 V, 0 V

4- Determine the voltage across the resistor 3 k Ω .



- a- 0.44V b- 0.9V c- 0.2V d- 0 V

5- To get a peak load voltage of 40V out of a bridge rectifier. What is the approximate rms value of secondary voltage?

- a) 0 V b) 14.4 V c) 28.3 V d) 56.6 V

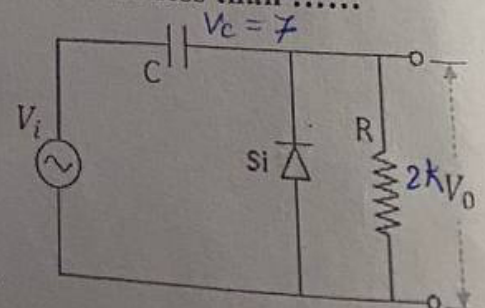
6- For a diode with $I_D = 25$ mA, the dynamic and DC resistance are

- a. $R_{\text{Dynamic}} = 31.6 \Omega$ and $R_{\text{DC}} = 31.6 \Omega$
b. $R_{\text{Dynamic}} = 31.6 \Omega$ and $R_{\text{DC}} = 25 \Omega$
c. $R_{\text{Dynamic}} = 1.04 \Omega$ and R_{DC} cannot be determined
d. R_{Dynamic} cannot be determined and $R_{\text{DC}} = 1.04$

7- In a full-wave rectifier circuit, if the output signal has a time period of 4 ms, then the input signal should have a frequency of

- a. 125 Hz b. 2 Hz c. 250 Hz d. 2 ms e. 250 KHz

8- To implement the following circuit as a clamper for $C = 7 \mu\text{F}$ and $R = 2 \text{ K}\Omega$ with assuming a simplified diode is used, the discharge period of the input signal should be less than

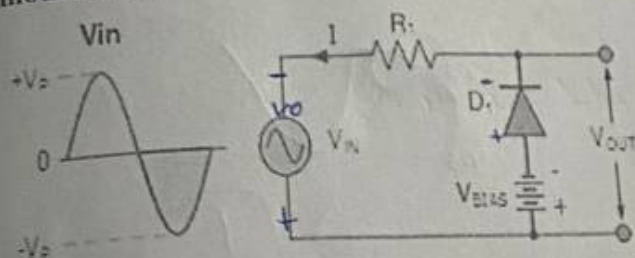


- a. 14 b. 28 ms c. 7 ms d. 70 ms e. None of them.

9- During the recombination process of $p-n$ junction, the use of Ge diode will

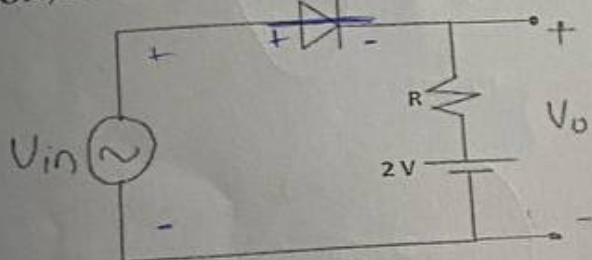
- a- Emit a light in invisible zone
- b- Emit light in the visible zone
- c- Dissipate the energy as a heat
- d- None of the above

10- The output voltage of the following circuit during the negative cycle is (Assume a simplified model of Si)



- a. $-V_p$
- b. $+V_p$
- c. $V_{bias} - 0.7\text{ V}$
- d. $-0.7 - V_{bias}$
- e. $V_p - 0.7$
- f. $-V_p + V_{bias}$

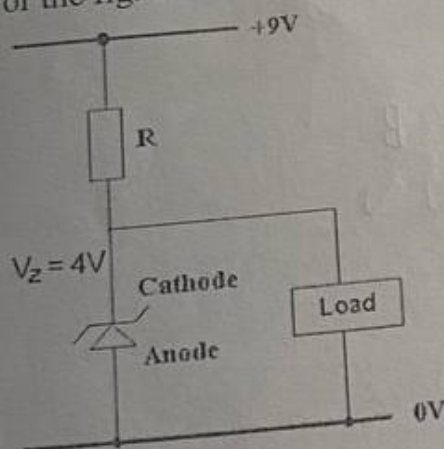
11- When the diode for the following circuit is an ideal diode, the output voltage V_o when the diode is ON, $R = 1\text{ K}\Omega$ is equal to



- a. $+2\text{ V}$
- b. V_{in}
- c. $2\text{ V} - 0.7\text{ V}$
- d. $0.7\text{ V} + V_R + 2\text{ V}$
- e. -2 V
- f. None of them

Problem 2 (4 points)

For the fig. below:



$$I_{min} = 10\text{ mA}$$

$$I_{max} = 90\text{ mA}$$

a- The minimum Zener diode current is 10 mA. If the maximum load current is 90 mA, calculate value for the series resistor R.

$$I_R = \frac{V_{in} - V_Z}{R}$$

$$R = \frac{9}{100\text{ mA}} = 9\text{ }\Omega$$

$$I_R = I_{Lmin} + I_{Zmax} = 100\text{ mA}$$

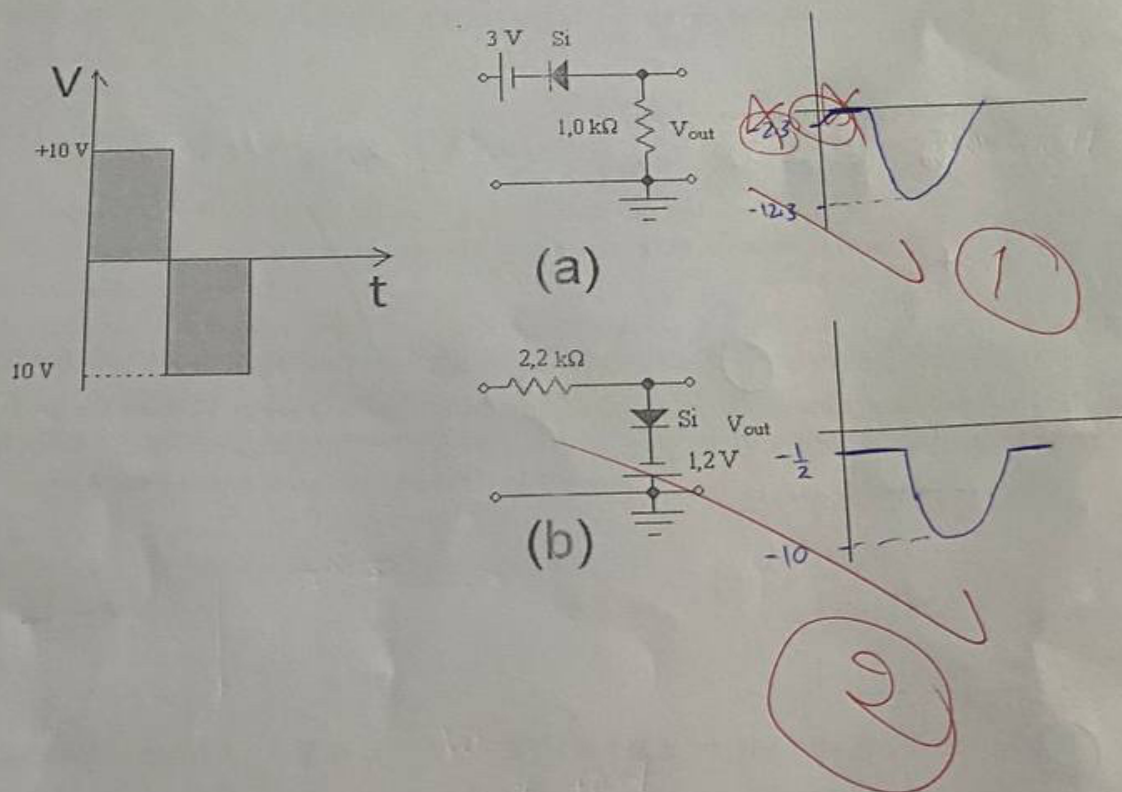
b- Calculate the power rating of the Zener diode in the diagram above.

$$P = IV$$

$$P = 0$$

Poroblem 3 (4 points)

Problem 4: For the input shown, determine the output voltage waveform for each circuit in figure 3(a, b, c), the diodes in this circuit practical.



Question	1	2	3	4	5	6	7	8	9	10	11
Answer	D	B	B	C	C	D	A	D	C	D	B

Good Luck